Joint	Fasteners	Torque / preload	Check
Blade root → pitch bearing inner ring	12 × M16 10.9 per blade	280 Nm (μ 0.12), re-torque at 50 h	F_bolt 64 kN design vs 113
Pitch bearing outer ring → hub neck	20 × M16 10.9 per blade	280 Nm	doc 35 §5
Hub → drum web	12 × M16 10.9 PCD 400	280 Nm	slip 1.7, separation 5.3
Spindle → bedplate	12 × M20 10.9 PCD 350	540 Nm	slip 6.4, separation 2.3
Stator carrier → spindle flange	12 × M12 10.9 PCD 240	110 Nm	slip 1.4
Disc → drum ring	24 × M12 10.9 PCD 662	110 Nm	slip 1.7
Caliper → bracket	4 × M16 10.9 each (3 calipers)	280 Nm	–
Third caliper bracket → bedplate	4 × M16 10.9 + nuts	280 Nm	–
Yaw inner ring → bedplate	16 × M16 10.9 HV PCD 350	280 Nm	util 0.42
Yaw outer ring → adapter	16 × M16 10.9 PCD 490	280 Nm	–
Adapter → tower top flange	16 × M20 10.9 HV PCD 380	540 Nm	ratio 0.45
Tower base flange → anchors	16 × M24 8.8 HDG, 3 nuts each	tension to 150 kN (tool), re-check at 3 months	F_bolt 202.5 kN
Hinge brackets → pedestal	4 × M24 8.8 HDG	350 Nm	–
Cover, spinner, mast, nose frame	M6–M12 A4-70	A4 table	–

NOTES

1. Torques for 10.9 zinc-flake with μ 0.12 (lubricated); mark every bolt after torquing; check 10 % of each joint at the first service (3 months) and yearly.
2. HV sets (EN 14399-4) where the joint carries the yaw moment; no washers under the nut side of HV sets other than those in the set.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

FASTENER AND TORQUE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-900	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked